

Name
Class
R. No.

• Mathematics • IIT JEE / CBSE
• Test 3 (July 20 , 2010) • Sets
• Maximum Marks 50 • Time 35 minutes

Instructions

1. The test consists of 3 sections making a total of 50 marks.

Section I

Q1: Write the set $\left\{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, \frac{6}{7}\right\}$ in the set-builder form.

6 Marks

Q2: How do you represent :

- i) B is a superset of A.
- ii) The set **T** of irrational numbers

4 Marks

Q3: Let $A = \{1, 2, \{3, 4\}, 5\}$. Which of the following statements are correct and which are incorrect.

- i) $\{3, 4\} \subset A$
- ii) $\{3, 4\} \in A$
- iii) $\{\{3, 4\}\} \subset A$
- iv) $1 \in A$
- v) $1 \subset A$
- vi) $\{1, 2, 5\} \subset A$
- vii) $\{1, 2, 5\} \in A$
- viii) $\phi \in A$
- ix) $\phi \subset A$
- x) $\{\phi\} \subset A$

5 Marks

Q4. Taking the set of natural numbers as the universal set, write down the complements of the following sets.

- i) $\{x: x \text{ is an even natural number}\}$
- ii) $\{x: x \text{ is an odd natural number}\}$
- iii) $\{x: x \text{ is a positive multiple of } 3.\}$
- iv) $\{x: x \text{ is a prime number.}\}$
- v) $\{x: x \text{ is a natural number divisible by } 3 \text{ and } 5\}$
- vi) $\{x: x \text{ is a perfect square}\}$
- vii) $\{x: x \text{ is a perfect cube}\}$
- viii) $\{x: 2x + 5 = 9\}$
- ix) $\{x: x \geq 7\}$
- x) $\{x: x \in \mathbb{N} \text{ and } 2x + 1 > 10\}$

10 Marks

Q5: Out of 500 car owners investigated , 400 owned car A and 200 owned car B, 50 owned both A and B cars. Is the data correct?

4 Marks

Q6: In a survey of 60 people , it was found that 25 people read newspapers H, 26 read newspaper T, 26 read newspaper I, 9 read both H and I, 11 read both H and T, 8 read both T and I, 3 read all three newspapers. Find :

- i) The number of people who read at least one of the newspapers
- ii) The number of people who read exactly one newspaper.

6 Marks

Section II

Q6: Show that A is a disjoint union of $A - B$ and $A \cap B$, i.e. show that

a) $A = (A - B) \cup (A \cap B),$

b) $(A - B) \cap (A \cap B) = \phi$

10 Marks

Section III

Q8: Determine the validity of the following argument:

S_1 : *Babies are illogical*

S_2 : *Nobody is despised who can manage a crocodile*

S_1 : *Illogical people are disliked*

S : *Babies cannot manage crocodiles*

5 Marks